Daniel Firoozabadi

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Education/Academic Summary

UC Berkeley - Mechanical Engineering/EECS - GPA: 3.814
Relevant Coursework: Introduction to Computer Programming for Scientists and Engineers, Three-Dimensional Modeling for Design, Structure and Interpretation of Computer Programs, Designing Information Devices and Systems I , Multivariable Calculus
Organizations: UC Berkeley Solar Vehicle Team, IEEE Berkeley Chapter, AI Ventures at Berkeley, Undergraduate Real Estate Investment Club, Pi Tau Sigma Engineering Honor Society
Credit Hours: 57.92, Expected Graduation: **Spring 2027**

High School Diploma, 05/2024 - Oak Park High School, CA - Summa Cum Laude with Highest Honors, 4.71 GPA, 1550 SAT
Relevant AP Course Summary: AP Physics 2, AP Statistics, AP Calculus BC, AP Computer Science Principles, AP Calculus AB, AP Biology
Dual Enrollment, 2023 - Moorpark College, CA - GPA: 4.0
Course Summary: Bus M30: Introduction to Business, Physics M10 - General Physics 1

Work/Research Experience

Software Engineering Intern, 01/2025 to present
Runopt Company
Optimized AI algorithms for civil site design to enhance cutting efficiency, reduce construction costs, accelerate design processes, and minimize environmental impact in real estate development:
- Developed an advanced 3D visualization framework using PyVista to improve design analysis and decision-making.
- Spearheaded the integration of OpenMDAO across multiple design modules, enabling scalable and efficient optimization for real-world applications.

Berkeley HAAS Entrepreneurship Hub - Startup Test Track Participant, 01/2025 to present
Exclusive program where I apply iterative, evidence-based entrepreneurship strategies to validate my startup idea through structured testing, customer feedback, and the Business Model Canvas.

Berkeley Skydeck ACE (Accelerating Careers in Entrepreneurship) Internship, 01/2025 to present
Selective Entrepreneurship Track where I learn from and connect with founders, venture capitalists, and prominent voices at UC Berkeley with weekly meetings and speaker events.

Susquehanna International Group Discovery Day Participant, 01/2025 - 01/2025
Competitive pre-internship program where I learned about quantitative trading, the role of global financial markets, as well as topics such as game theory and decision science.

Research Intern, 05/2023 to 08/2023
University of Southern California, USC - Viterbi Flying Light Specks Lab
Applied physics/engineering concepts in coding flight patterns for autonomous mini drones interfaced with Arduino at a graduate-level lab. I was designated as a contributor to the NSF award of $250,000 for the project “EAGER: Prototyping Touch with FLS-Matter” with Award ID: 2232382. Project Summary: https://github.com/dfirooza/off-the-shelf-drone-control

Student Researcher, 09/2024 to Present
UC Berkeley FHL Vive Center for Enhanced Reality
Immersive Semi-Autonomous Aerial Command System (ISAACS) Research Group; Computer Vision Subteam
Researching human-UAV interaction for transportation and mapping using a drone interfaced with Zed SDK. Our team is collaborating with the Lawrence Berkeley National Laboratory to perform 3D reconstruction of the environment through advanced methods in radiation detection:
- Designing and managing algorithms for object detection and depth camera calibration

Project Experience

Team Member, 08/2024 to Present
UC Berkeley Solar Vehicle Team (CalSol); Dynamics and Shell Subteams
Designing, manufacturing, and testing a full-powered solar vehicle for competition at the end of the year. As a part of the dynamics and shell subteams, I am working on the car’s suspension, steering, hubs, and brake systems, in addition to designing and manufacturing the shell of the solar vehicle:
- Designed lightweight and stable steering knuckle and centerlock using SolidWorks for Generation 11 Car
- Developed negative mold of updated car design
- Currently leading design of lightweight and robust car skid

Project Leader, 08/2024 to 12/24
UC Berkeley IEEE Micromouse (PCB Engineering)
- Engineered, built, and soldered an autonomous maze-solving robot (“mouse”) leveraging encoder odometry and I2C communication protocols.
- Developed and implemented a real-time A pathfinding algorithm* for optimized navigation and decision-making, achieving record maze-solving times.

Certifications

Python for Data Science with Real Exercises; Introduction to AWS; Natural Language Processing with Python - Udemy
MATLAB Onramp - MATLAB Coding; **Machine Learning: an overview** - Coursera
Fundamentals of Engineering at MIT - Summer Springboard
Youth Leadership Program of Conejo Valley - Conejo/Las Virgenes Future Foundation

Prior Activities & Awards

Math Club President: Led interactive meetings; provided key resources for competitions; Prepared students to take the AMC 10/12
STEM Club President and Founder: Organized group meetings and hands-on activities; recruited over 40 members
NHS Treasurer: 300+ members; managed club dues and budget; signed off on purchases; volunteered at monthly events
Science Olympiad Awards: 1st Place: Detector Building, 1st Place: Experimental Design, 2nd Place: Trajectory, 3rd Place: Fermi Questions (2x)
Future Business Leaders of America Awards: Intro to Business Procedures at Regional Conference - 1st place at sectional conference, 6th at State

Technical Skills

Languages/Libraries/Tools: Python, MATLAB, Javascript, HTML/CSS, Matplotlib, NumPy, VS Code, Pandas, PyVista
Data Focused: Natural Language Processing, Machine Learning, Statistical Data Analysis, AWS, COMSOL, OpenMDAO
Hardware/CAD: SolidWorks, Fusion360, Arduino, Soldering